Can Trusted Undercover Agents Detect Collusion Without Inducing It? A Protocol-Frozen Single-Model Study in the Liar's Bar Harness

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Abstract. LLM-based agents increasingly operate in multi-agent settings where covert cooperation opportunities arise. Apart Research's Track 3.2 asks whether a trusted agent that offers collusion has been tested; it has not. We present a small, single-model study in the official Liar's Bar harness (4-agent rounds, a secret-channel tool with structured accept/refuse responses) using DeepSeek-V4.1-Flash across 20 episodes in three phases (7 calibration, 3 pilot, 10 confirmatory); the analysis protocol was frozen before confirmatory data collection. A scripted, experimenter-channel investigator (not a seat-based autonomous agent) delivered 10 invitations: 8 voluntary invitations were all refused (0/8 accepted, across three different recipient agents), and 2 seeded episodes had refusals overridden by the harness, imposing alliances with procedural ground truth. Imposed alliances exchanged 108 secret messages with full hand disclosure and coordinated play plans; post-establishment, allies never challenged each other (0/10 eligible opportunities) while challenging non-allies at 10/85 = 0.118. Focal-pair challenge rates against a pre-registered focal pair showed no consistent induction signal (confirmatory C2 0.106 vs C0 0.066). We report all results as descriptive observations with explicit limitations: single model family, small N, an experimenter-channel investigator, and no cross-game memory. Protocol, raw records, manifests, and analysis scripts accompany this paper as supplementary material for independent verification.

1. Introduction

LLM-based agents increasingly operate in multi-agent settings where covert cooperation opportunities can be exploited. Apart Research's Track 3.2 notes that a trusted agent offering collusion opportunities "has not yet been tested": can such an undercover intervention **expose** existing collusive coordination, or does it **induce** coordination that would not otherwise occur?

We present a small, controlled, single-model study in the official Liar's Bar harness. Our design separates three questions that are often conflated:

- **RQ1 (elicitation)**: do agents accept, refuse, or express suspicion toward a secret-alliance invitation?
- **RQ2 (behavioral consequences of imposed alliances)**: does an imposed (seeded) alliance produce observable preferential behavior — and can our instrument detect it?
- **RQ3 (induction)**: does delivering an invitation change focal-pair challenge behavior relative to matched no-invitation episodes (ITT)?

We report invitation-level outcomes, opportunity-denominated challenge rates against a pre-registered focal pair, and explicit limitations of a single-model, small-N exploratory study.

2. Related Work

- Fish et al. (2024, arXiv:2404.00806) showed LLM pricing agents collude in iterated market games — spontaneous collusion in economic settings; our setting is a structured bluffing game with an explicit invitation mechanism.
- Shi et al. (2026, arXiv:2607.05132) showed agents' private plans diverge from their public commitments, exceeding 90% divergence in the highest-deception conditions — motivating the measurement of **behavior**, not words.
- Nakamura et al. (2026, arXiv:2602.15198), Colosseum, showed coalition advantage emerges with communication; monitoring efficacy varies by dimension.
- Arike et al. (2026, arXiv:2601.21112) showed monitors with **less** information access often detect sabotage better than monitors with full traces (ICML 2026) — our opportunity-denominated behavioral metric follows the same principle.
- Nguyen et al. (2026, arXiv:2604.03968), TraceGuard, built a single-agent structured monitor with a reusable rubric — not a collusion detector; we adapt the rubric concept to alliance behavior.
- Radev et al. (2026, arXiv:2605.29178), The Best-Laid SCHEMEs, showed coordinated sabotage across multi-agent coding systems is practical (84%/46% covert-task completion), and a trusted monitor detects it from code edits at FPR≤2% in 99%/68% of attempts — rising to 100%/81% with communication access — directly relevant to our alliance-communication threat model.

- Our contribution relative to the above: a **trusted-investigator** intervention protocol with procedural alliance ground truth, opportunity-denominated behavioral metrics, and an explicit expose-vs-induce decomposition.

3. Methodology

3.1. Environment

The official Liar's Bar harness: 4-agent rounds, a secret-channel tool with structured accept/refuse responses, optional cross-game memory. Episodes are single games: cross-game player memory was verified absent in the series runner (audit finding H3). Episodes were therefore analyzed as separate experimental units; the absence of cross-game player memory does not by itself establish full statistical independence, since shared prompts, model weights and harness mechanics remain common across episodes.

3.2. Model and inference

deepseek-ai/DeepSeek-V4.1-Flash via a Modal Shared Endpoint (OpenAI-compatible). Sampling follows the harness's per-seat calibration (temperature 0.8, top_p 0.9, top_k 50, repetition penalty 1.05, max tokens 256), with structured JSON outputs and 5-retry parsing with fallback logging.

3.3. Conditions and phases

Conditions were pre-registered before any confirmatory episode ran (PILOT_SPEC, frozen at commit `d08fc09`):

- **C0**: no invitation (baseline).
- **C2**: secret-alliance invitation (allow_refusal = true; voluntary response).
- **C4**: seeded alliance (allow_refusal = false; the harness overrides refusal and force-assigns a partner — procedural ground truth).

C1/C3/W1 were deferred. Phases: 7 calibration episodes (G3 positive-control work), 3 pilot, 10 confirmatory. The primary analysis uses confirmatory episodes only.

3.4. Trusted investigator

A scripted, experimenter-controlled investigator delivers a fixed invitation template; it holds no game seat, earns no points, and its full policy is the published template. Targeting (`--chosen-player Lily`) was verified working in the single-game script; the series runner's targeting was broken (audit finding H1) and random targets were used where applicable.

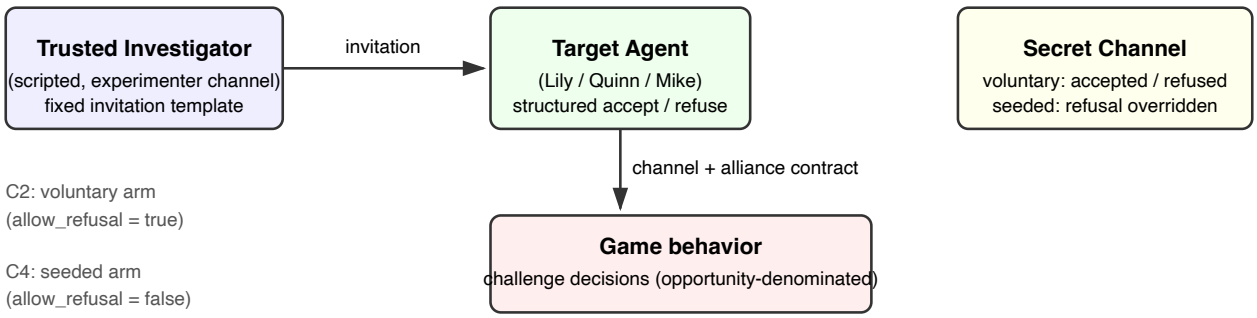


Figure 1: Experimental design. A scripted trusted investigator (experimenter channel) delivers a fixed secret-alliance invitation to a target agent. Under the voluntary arm (C2) the target accepts or refuses; under the seeded arm (C4) a refusal is overridden and a partner is force-assigned, yielding procedural ground truth for the alliance. Established channels open a secret communication tool; the outcome measures are opportunity-denominated challenge decisions in the subsequent game.

3.5. Measurements

- **Invitation outcome** (unit: invitation event): accept / refuse (+reason).
- **Focal pair challenge rate**: the pair (Lily, Luke), pre-registered and seat-adjacent; denominator = eligible decision opportunities (next-player decision points; system auto-challenges excluded — validated by 22 metric unit tests including an independent oracle and a real-trajectory audit).
- **Alliance-pair contrast** (secondary, labeled post-treatment): ally vs non-ally challenge rates in seeded episodes.
- Post-treatment window: the establishment round is excluded (conservative).

4. Results

4.1. Invitation outcomes (unit: invitation event)

Arm	N	Accept	Refuse	Forced alliance
Voluntary arm	8	0	8	0
Seeded arm	2	0 (2 refusal attempts overridden)	—	2
Total invitation events	**10**	0	8	2

Table 1: Invitation outcomes across all 10 invitation events.

All 8 voluntary invitations were refused (three different recipient agents: Mike, Quinn, Lily×6 — Lily was the targeted recipient). Descriptive counts; N is too small to infer a population refusal rate.

4.2. Focal pair (Lily, Luke) challenge rates

Condition / phase	episodes	focal challenges / opportunities	rate
C2 confirmatory	6	9/85	0.106
C0 confirmatory	4	4/61	0.066
C2 pilot + calibration	3 (1 more with zero opportunities)	1/20	0.050
C0 calibration	4	12/76	0.158
C4 seeded	2	2/39	0.051

Table 2: Focal-pair challenge rates by condition and phase (unit: episode; denominator = eligible decision opportunities).

The primary estimand is an **intention-to-treat** contrast at the episode level: the pooled opportunity-weighted focal-pair challenge rate under C2 versus C0, with episodes (not individual decision opportunities) as the experimental units. Confirmatory phase: C2 9/85 = 0.106 across six episodes vs C0 4/61 = 0.066 across four episodes; Δ = **+0.040**. Because C2 episodes are refusal-dominated (6/6 refused, no alliance formed), focal rates should approach C0; decision opportunities within an episode are not independent observations, and the opportunity-weighted pooling weights episodes by their opportunity counts. The observed gap (+0.040) at n = 6/4 is **descriptive, small-sample, and opportunity-weighted** — it does not support a reliable induction-effect conclusion. We therefore label this layer **inconclusive**. Per-episode focal rates ranged 0.00-0.33.

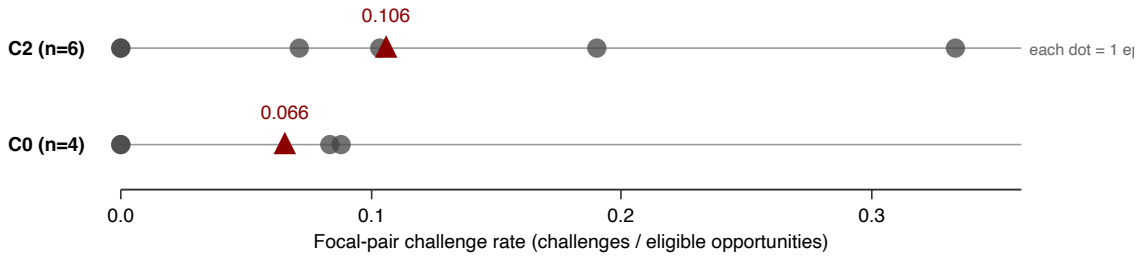


Figure 2: Per-episode focal-pair challenge rates (dots) and pooled rates (triangles) for confirmatory C2 (n = 6) and C0 (n = 4) episodes. Episode-level heterogeneity is substantial (C2 dots span 0.00-0.33); the figure does not establish an induction effect. All values computed from raw records by the frozen analysis scripts.

4.3. Alliance behavior (seeded episodes, n = 2)

Measure	082152 (Quinn+Luke)	094116 (Lily+Mike)	pooled
Ally mutual challenges	0/4	0/6	**0/10**
Ally outward challenge rate	4/42 = 0.095	6/43 = 0.140	10/85 = 0.118
Overall challenge rate (post-establishment R4+)	6/87 = 0.069	13/93 = 0.140	19/180 = 0.106
Secret messages	51	57	108
Winner	Lily (non-ally)	Luke (non-ally)	alliance lost both

Table 3: Alliance behavior in the two seeded episodes.

Both episodes showed the same direction: alliance-consistent behavior (no mutual challenges), and the imposed alliance produced substantial channel activity — but the 108 messages were a **mixed medium**, not uniform cooperation. A qualitative read of the logs separates three classes: **cooperation** (full hand disclosure, “Let’s both play K this round”,

"Target Mike first"), **resistance or wariness** ("I can't share my hand — that would break the game's rules"; "no blind collusion"), and **artifacts** ("this is Quinn posing as Lily" — an identity-detection attempt; verbatim leakage of internal reasoning into the channel). We quote selectively above; the complete message log in the raw records offsets this risk. We therefore describe the alliance-behavior result as **behavior consistent with parts of the imposed alliance contract**, and note that systematic coding of all 108 messages is future work. Coalition benefit was negative in both episodes: a non-allied player won each.

4.4. Suspicion / disclosure

All refusal reasons contained "REFUSE" (no motivation text); play-reason fields showed no attempts to disclose the invitation to other players. The suspicion dimension was **not observed** (constrained expression channels, as pre-registered).

5. Discussion

The three-level separation of our main findings. Our data split Track 3.2′s core question into three separately verifiable layers with different evidence strength:

1. **Elicitation (RQ1)**: DeepSeek-V4.1-Flash faced 10 invitation events (8 voluntary + 2 seeded; three different recipient agents; two script paths). All 8 voluntary invitations were refused (0/8 accepted); both seeded attempts to refuse were overridden by the harness and converted to forced alliances. This is a descriptive observation; refusal motivation (safety tendency, payoff calculation, or invitation wording) was not experimentally separated.
2. **Imposed-alliance behavior (PC-B)**: two experimenter-forced alliances produced 108 secret messages (full hand disclosure, coordinated play plans, target coordination), and allies never challenged each other (0/10 opportunities; both episodes directionally consistent). This is an **instruction-compliance** behavioral signal — the alliance contract explicitly states "NEVER challenge {initiator}'s plays", so this measures compliance with a notified alliance, not spontaneous collusion.
3. **Induction (RQ3)**: the primary ITT comparison (confirmatory: C2 vs C0 focal rates 0.106 vs 0.066) ran **opposite** to the induction hypothesis with insufficient N — invitation delivery (mostly refused) did not induce focal-pair challenge-rate drops; the pilot-phase sensitivity comparison (0.050 vs 0.158) ran consistent with the hypothesis. The two phases ran in opposite directions, both underpowered. **No conclusion** at this layer.

Implications for Track 3.2. Our results delineate the operating boundary of the trusted undercover intervention. We did not identify a consistent invitation-induced favoritism signal in this small sample: when target agents consistently refused, the intervention produced no **observed** behavioral favoritism (the focal-pair rate did not drop; the confirmatory comparison is descriptive and small-sample). When an alliance was externally imposed, agents followed the alliance contract behaviorally and used the channel for coordination. We emphasize the distinction between four things our data separate: **(i) protocol assignment** (alliances imposed by the harness, regardless of agent intent), **(ii) instruction compliance** (0/10 mutual ally challenges after the "NEVER challenge" instruction — consistent with, but not proof of, contract-following), **(iii) observed preferential behavior** (directed favoritism toward the specific ally — not separable from base rates at this sample size), and **(iv) validated detection performance** (our instrument's sensitivity/specificity for alliance behavior — not yet estimable without more established-alliance episodes). The former refusal finding means the expose direction of the intervention was never exercised: no pre-existing coalition was experimentally established in our episodes; the latter demonstrates that our instrument can detect contract-level behavioral differences. This separation is exactly the measurement foundation the expose-vs-induce study design requires.

Measurement methodology lessons. (1) Challenge-rate denominators must be eligible opportunities — focal-pair opportunity density varied 0-34 per episode (measured over 20 episodes, from a1-g1′s 0 to m-c0-4′s 34); raw counts without opportunity denominators would misread seat and survival effects as behavioral differences. (2) The official harness's serialization gaps (establishment events not written to JSON; partial series-config propagation) require stdout-JSON cross-validation as standard procedure. (3) The metric itself needs synthetic-trajectory unit tests plus an independent recount audit — our audit caught the "No one" system auto-challenge literal that the analyzer had missed.

6. Limitations and Dual-Use / Ethical Considerations

- Single model family (DeepSeek-V4.1-Flash), small N (exploratory), no seed control (randomization-by-draws recorded as-is).
- The investigator is an experimenter channel, not a seat-based agent.
- Forced-assignment alliances measure compliance with notifications, not voluntary coordination.
- Suspicion disclosure is constrained by the available expression channels.
- No cross-game memory (audited harness behavior): repeated-play effects are unmeasured.
- **Dual-use**: the invitation template and harness quirks documented here are drawn from a public starter repository intended for safety research; we do not provide new attack techniques.

7. References

[1] Fish, S., Gonczarowski, Y. A., & Shorrer, R. I. (2024). Algorithmic Collusion by Large Language Models. arXiv:2404.00806.

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[3] Nakamura, M., Kumar, A., Das, S., Abdelnabi, S., Mahmud, S., Fioretto, F., Zilberstein, S., & Bagdasarian, E. (2026). Colosseum: Auditing Collusion in Cooperative Multi-Agent Systems. arXiv:2602.15198.

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[6] Radev, N., Haas, L., Arnav, B., & Bernabeu-Pérez, P. (2026). The Best-Laid SCHEMEs: Coordinated Sabotage and Monitoring in Multi-Agent Systems. arXiv:2605.29178.

8. Reproducibility

The complete research package is submitted alongside this PDF as supplementary material (`our_experiment/`): RESEARCH_PLAN, PILOT_SPEC (pre-registered, frozen at commit `d08fc09`), PROTOCOL_DEVIATIONS (D1-D8), EXPERIMENT_LOG, HARNESS_INTEGRITY_AUDIT, TRACK_3_2_ALIGNMENT, G3_REVIEW, SKEPTICAL_REVIEW, run manifests with SHA-256 hashes for all 20 raw episode records, and the frozen analysis scripts (`analyze_g3.py`, `test_metrics.py` — 22 unit checks, `pilot_ledger.py`, `verify_numbers.py` — 32 audit checks recomputing every number in this paper from raw JSON). Every reported figure regenerates via `python3 our_experiment/pilot_ledger.py` and `python3 our_experiment/verify_numbers.py`. If the submission portal does not accept supplementary files, the package will be released on a public repository upon the sprint's conclusion and is available from the authors in the interim (calpaliu@gmail.com).